

Friction & Surface Type

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

Cue focus	Friction acts opposite relative motion; rougher surfaces usually produce more friction and smoother surfaces usually produce less.
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Round 1 - Retrieve It

1. Which direction does friction act compared with motion?

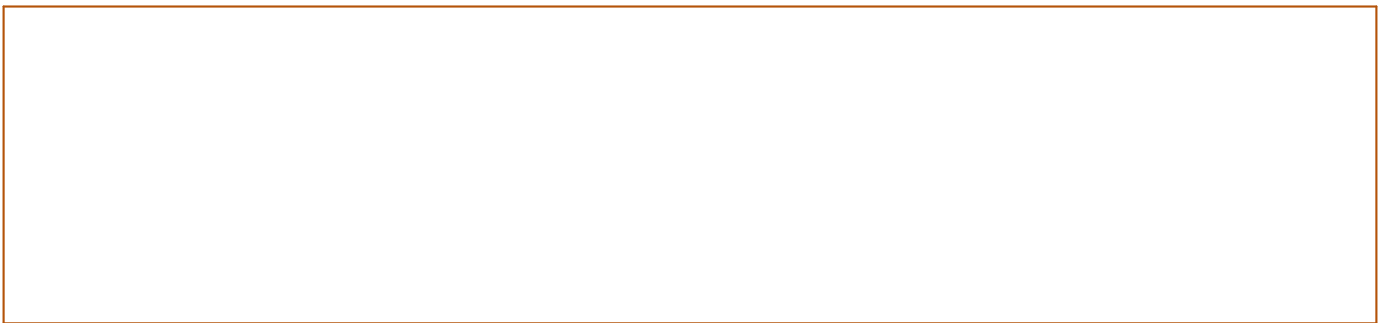
2. Which usually produces more friction: a rough surface or a smooth surface?

3. Give one situation where more friction is helpful.

4. Give one situation where less friction is helpful.

Round 2 - Sketch It

Draw the same block moving right on a smooth surface and a rough surface. Add friction arrows and compare their relative sizes.



Friction & Surface Type - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 - Apply the Relationship

1. Rank likely friction from least to greatest: ice, smooth tile, carpet, sandpaper.

2. Why do athletic shoes often have textured soles?

3. Why does adding lubricant to machine parts reduce heating and wear?

Round 4 - Reason from Evidence

Friction can be both useful and harmful. Explain one example of each using force and motion language.

Friction & Surface Type - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain Friction & Surface Type in your own words. Use one example that is not printed on the cue card.

C**Connect**

Connect friction to net force. If friction is the only horizontal force on a moving object, what happens to its motion?

E**Extend**

Design a fair test to compare friction on two surface types. Identify what you would keep the same and what you would measure.

Confidence check Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

TEACHER KEY - Friction & Surface Type

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Opposite the direction of relative motion or attempted motion.
2. Rough surface.
3. Examples: shoe traction, tires gripping, brakes stopping, holding objects.
4. Examples: lubricated machine parts, skating, sliding an object more easily.

Sketch It - look for

Friction arrows point left. Rough-surface arrow should be larger than smooth-surface arrow for the same setup.

Apply It

1. Ice, smooth tile, carpet, sandpaper (general expectation).
2. To increase friction/traction with the ground.
3. It reduces friction between moving surfaces.

Reason from Evidence - look for

Helpful example should show friction providing needed grip/control. Harmful example should show friction opposing motion, causing wear, or converting useful mechanical energy to thermal energy.

ACE - Extend look-for

Keep object, mass, pulling method/angle, and distance constant; change surface type; measure force needed or stopping distance/time.

Suggested use

Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.